DIFFERENCE EQUATIONS

Many problems in Probability give rise to difference equations. Difference equations relate to differential equations as discrete mathematics relates to continuous mathematics. Anyone who has made a study of differential equations will know that even supposedly elementary examples can be hard to solve. By contrast, elementary difference equations are relatively easy to deal with.

Aside from Probability, Computer Scientists take an interest in difference equations for a number of reasons. For example, difference equations frequently arise when determining the cost of an algorithm in big-O notation. Since difference equations are readily handled by program, a standard approach to solving a nasty differential equation is to convert it to an approximately equivalent difference equation.

Recall:

Given a set of values

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$$

satisfying $y = f(x)$, with

$$x_i = x_0 + ih,\ h > 0,\ i = 1, 2, \ldots, n.$$

the differences

$$y_1 - y_0,\ y_2 - y_1, \ldots, y_n - y_{n-1}$$

are called first order forward differences of y and are respectively denoted by $\Delta y_0, \Delta y_1, \Delta y_2, \ldots, \Delta y_n$. We call 'Δ', the forward difference operator. The differences of the first order forward differences are called second order forward differences and are denoted by $\Delta^2 y_0,\ \Delta^2 y_1,\ \Delta^2 y_2,\ \ldots,\ \Delta^2 y_{n-1}$. Thus

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0 = y_2 - 2y_1 + y_0, \quad \Delta^2 y_1 = \Delta y_2 - \Delta y_1 = y_3 - 2y_2 + y_1$$

and so on. Similarly, one can define the r^{th} order forward differences recursively as,

$$\Delta^r y_k = \Delta^{r-1} y_{k+1} - \Delta^{r-1} y_k,\ r = 1, 2, \ldots$$

Shift Operator:

The shift operator, denoted by E, is defined by the equation

$$Ey_k = y_{k+1}.$$

The operator shifts the functional value y_k to the next higher value y_{k+1}.

Again operating by E, we get

$$E^2 y_k = E(Ey_k) = E(y_{k+1}) = y_{k+2}.$$

In general

$$E^r y_k = y_{k+r}.$$

Definition: A difference equation is a relation between the differences of an unknown function at one or more general values of the argument.

Examples:

$$\Delta y_{n+1} + y_n = 2 \quad \ldots\ldots. (1)$$

$$\Delta y_{n+1} + \Delta^2 y_{n-1} = 1 \quad \ldots\ldots\ldots.. (2)$$

Since $\Delta y_{n+1} = y_{n+2} - y_{n+1}$, $\Delta^2 y_{n-1} = y_{n+1} - 2y_n + y_{n-1}$, (1) and (2) can be written as

$$y_{n+2} - y_{n+1} + y_n = 2 \quad \ldots\ldots\ldots\ldots. (3)$$

$$y_{n+2} - 2y_n + y_{n-1} = 1 \ \ldots\ldots.. (4)$$

Order of a Difference Equation is the difference between the largest and the smallest arguments occurring in the difference equation divided by the unit of increment.

For example,

(3) above is the second order, for

$$\frac{\text{largest argument} - \text{smallest argument}}{\text{unit of increment}} = \frac{(n+2) - n}{1} = 2$$

and (4) is of the third order, for

$$\frac{(n+2)-(n-1)}{1} = 3$$

Note: While finding the order of a difference equation, it must always be expressed in a form free of the operator Δ.

Solution of a difference equation is an expression for y_n which satisfies the given difference equation.

The **general solution** of a difference equation is that in which the number of arbitrary constants is equal to the order of the difference equation.

A **particular solution** or **particular integral** is that solution which is obtained from the general solution by giving particular values to the constants.

Formation of Difference Equations:

Example 1:- From the difference equation corresponding to the family of curves $y = ax + bx^2$

Solution: We have

$$y = ax + bx^2 \quad \ldots.(i)$$

$$\Delta y = a\Delta(x) + b\Delta(x^2) = a(x + 1 - x) + b[(x+1)^2 - x^2]$$

$$= a + b(2x + 1) \quad \ldots\ldots.. (ii)$$

$$\Delta^2 y = 2b[(x+1) - x] = 2b \quad \ldots\ldots.. (iii)$$

To eliminate a and b, we have

from (iii), $b = ½\, \Delta^2 y$

and from (ii), $a = \Delta y - b(2x+1) = \Delta y - \frac{1}{2}\Delta^2 y(2x+1)$

Substituting these values of a and b in (i), we get

$$y = \left[\Delta y - \frac{1}{2}\Delta^2 y(2x+1)\right]x + \frac{1}{2}\Delta^2 y . x^2$$

Or $\quad (x^2 + x)\Delta^2 y - 2x\,\Delta y + 2y = 0$

This is the desired difference equation which can also be written as

$$(x^2 + x)y_{x+2} - (2x^2 + 4x)y_{x+1} + (x^2 + 3x + 2)y_x = 0$$

Example 2: From $y_n = A2^n + b(-3)^n$, obtain difference equation by eliminating the constants.

Solution:

We have

$$y_n = A \,.\, 2^n + b(-3)^n,$$

$$y_{n+1} = 2A \,.2^n - 3b\,(-3)^n \quad \text{and}$$

$$y_{n+2} = 4A \,.\, 2^n + 9B\,(-3)^n$$

Eliminating A and B, from the above three equations, we get

$$\begin{vmatrix} y_n & 1 & 1 \\ y_{n+1} & 2 & -3 \\ y_{n+2} & 4 & 9 \end{vmatrix} = 0 \text{ i.e., } y_{n+2} + y_{n+1} - 6y_n = 0$$

Exercises:

1. Find the difference equation satisfied by $y = ax^2 - bx$
2. Derive the difference equations in each of the following cases:

   i) $y_n = A.3^n + B.5^n$ $\qquad$ ii) $y_x = (a + bx)2^x$

Linear Difference Equations

Definition: A linear difference equation is that in which y_{n+1}, y_{n+2}, etc occur in first degree only and are not multiplied together.

A linear difference equation with constant coefficient is of the form

$$y_{n+r} + a_1 y_{n+r-1} + a_2 y_{n+r-2} + \ldots\ldots + a_r y_n = f(n) \quad \ldots\ldots\ldots\ldots (1)$$

where $a_1, a_2, \ldots. a_r$ are constants.

If $u_1(n), u_2(n), \ldots\ldots. u_r(n)$ are r independent solutions of the equation

$$y_{n+r} + a_1 y_{n+r-1} + \ldots. + a_r y_n = 0 \quad \ldots\ldots\ldots.(2)$$

then, its solution is

$$u_n = c_1 u_1(n) + \ldots.. + c_r u_r(n) \qquad \ldots\ldots\ldots\ldots\ldots. (3)$$

where $c_1, c_2, \ldots., c_r$ are arbitrary constants.

If v_n is a particular solution of (1), then the complete solution of(1) is

$$y_n = u_n + v_n.$$

The part u_n is called the complementary function (CF) and the part v_n is called the particular integral (P.I) of (1).

Thus the complete solution of (1) is

$$y_n = CF + PI$$

Rules for finding Complementary Function

Consider

$$y_{n+r} + a_1 y_{n+r} + \dots + a_r y_n = 0$$

i.e, $(E^r + a_1 E^{r-1} + \dots + a_r) y_n = 0$

The auxiliary equation is

$$m^r + a_1 m^{r-1} + \dots + a_r = 0.$$

Let $m_1, m_2, \dots, m_r$ be the roots.

Case 1:- Suppose that all roots are real and distinct. Then the solution is

$$y_n = c_1 m_1^n + c_2 m_2^n + \dots + c_r m_r^n$$

Case 2:- If two roots are real and equal, say $m_1 = m_2$, and all other roots real and distinct, then

$$y_n = (c_1 + c_2 n) m_1^n + c_3 m_3^n + \dots + c_r m_r^n$$

If $m_1 = m_2 = m_3$, then

$$y_n = (c_1 + c_2 n + c_3 n^2) m_1^n + c_4 m_4^n + \dots + c_r m_r^n$$

Case 3:- Suppose that two roots are complex, say**,** $m_1 = \alpha + i\beta$, $m_2 = \alpha - i\beta$

then, the solution is

$$y_n = r^n (c_1 \cos n\theta + c_2 \sin n\theta) + c_3 m_3^n + \dots + c_r m_r^n \text{ where } r = \sqrt{\alpha^2 + \beta^2}, \theta = \sin^{-1}\left(\frac{\beta}{\alpha}\right)$$

Example 1: Solve the difference equation $u_{n+3} - 2u_{n+2} - 5u_{n+1} + 6u_n = 0$

Solution: Given equation in symbolic form is

$$(E^3 - 2E^2 - 5E + 6) u_n = 0$$

$\therefore$ Its auxiliary equation is

$$m^3 - 2m^2 - 5m + 6 = 0$$

$$(m - 1)(m + 2)(m - 3) = 0$$

Roots are 1, - 2, 3.

Thus the solution is $u_n = c_1(1)^n + c_2(-2)^n + c_3(3)^n$

Example 2:- Solve $u_{n+2} - 2u_{n+1} + u_n = 0$

Solution: Given equation in symbolic form is $(E^2 - 2E + 1) u_n = 0$

$\therefore$ Its auxiliary equation is $m^2 - 2m + 1 = 0$ Or $(E - 1)^2 = 0$.

Roots are 1, 1.

Thus the required solution is

$u_n = (c_1 + c_2 n)(1)^n$, i.e. $u_n = c_1 + c_2 n$

Example 3:- Solve $y_{n+1} - 2y_n \cos \alpha + y_{n-1} = 0$

Solution:-

$(E^2 - 2E \cos \alpha + 1)y_{n-1} = 0$

$\therefore$ Its auxiliary equation is $m^2 - 2m\cos \alpha + 1 = 0$

$$m = \frac{2 \cos \alpha \pm \sqrt{4 \cos^2 \alpha - 4}}{4} = \cos \alpha \pm i \sin \alpha$$

Thus the solution is

$y_{n-1} = (1)^{n-1} [c_1 \cos (n - 1)\alpha + c_2 \sin (n - 1)\alpha]$

Or $y_n = c_1 \cos n\alpha + c_2 \sin n\alpha$

Example 4:- The integers 0, 1, 1, 2, 3, 5, 8, 13, 21,..., are said to form a Fibonacci sequence. Form the Fibonacci difference equation and solve it.

Solution: In this sequence, each number beyond the second, is the sum of its two previous number. If y_n be the n^{th} number, then $y_n = y_{n-1} + y_{n-2}$ for $n > 2$.

Or $y_{n+2} - y_{n+1} - y_n = 0$ (for $n > 0$)

$(E^2 - E - 1)\, y_n = 0$ is the difference equation.

It's A.E is $m^2 - m - 1 = 0$ which gives $m = ½ (1 \pm \sqrt{5})$

Thus the solution is

$$y_n = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n, \text{for } n = 0, 1, 2, 3,$$

$y_0 = 0$ gives $c_1 + c_2 = 0$.

$y_1 = 1$ gives $1 = c_1 \frac{1+\sqrt{5}}{2} + c_2 \frac{1-\sqrt{5}}{2}$

Solving we get $c_1 = 1/\sqrt{5}$, $c_2 = -1/\sqrt{5}$.

Hence the solution (or nth Fibonacci number) is

$$y_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$$

Exercise:

Solve the following difference equations:-

1. $u_{x+2} - 4\, u_{x+1} + 4\, u_x = 0$ given $u_0 = 1$, $u_1 = 0$
2. $\Delta^2 u_n + 2\Delta u_n + u_n = 0$
3. $(\Delta^2 + 3\Delta + 2)\, y_n = 0$
4. $y_{n+3} - 3\, y_{n+2} + 3y_{n+1} - y_n = 0$

Rules for finding the Particular Integral:-

Consider equation

$$y_{n+r} + a_1 y_{n+r-1} + \dots + a_r y_n = f(n),$$

which in symbolic form is

$$\phi(E)\, y_n = f(n), \text{ where } \phi(E) = E^r + a_1 E^{r-1} + \dots + a_r$$

Then the particular integral is given by $\text{P.I.} = \frac{1}{\phi(E)} f(n)$

Case I:- When $f(n) = a^n$

$$\text{P.I.} = \frac{1}{\phi(E)} a^n,$$

$$= \frac{1}{\phi(a)} a^n, \text{provided } \phi(a) \neq 0$$

If $\phi(a) = 0$, then for the equation

i) $(E - a)\, y_n = a^n$, $\text{P.I} = \frac{1}{E-a} a^n = n a^{n-1}$

ii) $(E - a)^2\, y_n = a^n$, $\text{P.I} = \frac{1}{(E-a)^2} a^n = \frac{n(n-1)}{2!} a^{n-2}$

iii) $(E - a)^3\, y_n = a^n$, $\text{P.I} = \frac{1}{(E-a)^3} a^n = \frac{n(n-1)(n-2)}{3!} a^{n-3}$

and so on

Example 1:- Solve $y_{n+2} - 4y_{n+1} + 3y_n = 5^n$

Solution:-

Given equation in symbolic form is $(E^2 - 4E + 3)y_n = 5^n$

$\therefore$ The auxiliary equation is $m^2 - 4m + 3 = 0$

Roots are 1, 3

$\therefore$ C.F = $c_1(1)^n + c_2(3)^n = c_1 + c_2 \,.\, 3^n$

And

$$\text{P.I.} = \frac{1}{E^2 - 4E + 3} 5^n \quad [\text{put } E = 5]$$

$$= \frac{1}{25 - 4.5 + 3} 5^n = \frac{1}{8} . 5^n$$

Thus the complete solution is $y_n = c_1 + c_2 .3^n + 5^n/8$

Example 2:- Solve $u_{n+2} - 4u_{n+1} + 4y_n = 2^n$

Solution:-

Given equation in symbolic form is $(E^2 - 4E + 4)u_n = 2^n$

$\therefore$ The auxiliary equation is $m^2 - 4m + 4 = 0$

$\therefore$ Roots are 2, 2

$\therefore$ C.F = $(c_1 + c_2 n)2^n$

$$\text{P.I.} = \frac{1}{(E-2)^2} . 2^n$$

$$= \frac{n(n-1)}{2!} . 2^{n-2} = n(n-1)2^{n-3}$$

Thus the complete solution is

$$u_n = (c_1 + c_2 n)2^n + n(n-1)\ 2^{n-3}$$

Case II: (i) When f(n) = sinkn

$$P.I = \frac{1}{\phi(E)} sinkn$$

$$= \frac{1}{\phi(E)}\left(\frac{e^{ikn} - e^{-ikn}}{2i}\right)$$

$$= \frac{1}{2i}\left[\frac{1}{\phi(E)} a^n - \frac{1}{\phi(E)} b^n\right], \text{ where } a = e^{ik} \text{ and } b = e^{-ik}$$

Now proceed as in case I.

(ii) when f(n) = coskn,

$$\text{P.I.} = \frac{1}{\phi(\text{E})} \text{coskn}$$

$$= \frac{1}{\phi(\text{E})}\left(\frac{\text{e}^{\text{ikn}} + \text{e}^{-\text{ikn}}}{2}\right)$$

$$= \frac{1}{2}\left[\frac{1}{\phi(\text{E})} \text{a}^{\text{n}} + \frac{1}{\phi(\text{E})} \text{b}^{\text{n}}\right], \text{ as before.}$$

Now proceed as in case I

Example 3:- Solve $y_{n+2} - 2\cos\alpha \,.\, y_{n+1} + y_n = \cos n\alpha$

Given equation in symbolic form is

$$E^2 - 2\cos\alpha \,.\, E + 1 = 0$$

$$\therefore\ m = \frac{2\cos\alpha \pm \sqrt{(4\cos^2\alpha - 4)}}{2} = \cos\alpha \pm i\sin\alpha$$

$$r = \sqrt{cos^2\alpha + sin^2\alpha} = 1,\ \theta = tan^{-1}\left(\frac{sin\alpha}{cos\alpha}\right) = \alpha$$

$\therefore$ C.F = $(1)^n$ [c_1 cosnα + c_2 sinnα] i.e. CF = $c_1 \cos\alpha n + c_2 \sin\alpha n$

$$\text{P.I.} = \frac{1}{E^2 - 2E\,cos\alpha + 1} \cos\alpha n$$

$$= \frac{1}{E^2 - E(e^{i\alpha} + e^{-i\alpha}) + 1}\left(\frac{e^{i\alpha n} + e^{-i\alpha n}}{2}\right)$$

$$= \frac{1}{2}\left[\frac{1}{(E-e^{i\alpha})(E-e^{-i\alpha})} e^{i\alpha n} + \frac{1}{(E-e^{i\alpha})(E-e^{-i\alpha})} e^{-i\alpha n}\right]$$

$$= \frac{1}{2}\left[\frac{1}{(E-e^{i\alpha})} \cdot \frac{1}{e^{i\alpha} - e^{-i\alpha}} e^{i\alpha n} + \frac{1}{E-e^{-i\alpha}} \cdot \frac{1}{e^{-i\alpha} - e^{i\alpha}} e^{-i\alpha n}\right],\ e^{i\alpha} - e^{-i\alpha} = 2i\sin\alpha$$

$$= \frac{1}{4isin\alpha}\left[\frac{1}{E-e^{i\alpha}} e^{i\alpha n} - \frac{1}{E-e^{-i\alpha}} e^{-i\alpha n}\right] = \frac{1}{4isin\alpha}\left[n.e^{i\alpha(n-1)} - ne^{-i\alpha(n-1)}\right]$$

$$= \frac{n}{2\sin\alpha}\left[\frac{e^{i\alpha(n-1)} - e^{-i\alpha(n-1)}}{2i}\right] = \frac{n\sin(n-1)\alpha}{2\sin\alpha}$$

Hence the complete solution is

$$y_n = c_1 \cos \alpha n + c_2 \sin \alpha n + \frac{n \sin(n-1)\alpha}{2 \sin\alpha}$$

Case III:- when $f(n) = n^p$,

$$P.I = \frac{1}{\phi(E)} n^p = \frac{1}{\phi(1+\Delta)} n^p$$

1) Expand $[\phi(1 + \Delta)]^{-1}$ in ascending powers of Δ by binomial theorem as far as the term in Δ^p.

2) Write n^p in the factorial form and operate on it with each term of the expansion.

Note: While expanding in ascending powers, the following formulae are useful.

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots$$

$$\frac{1}{(1-z)^2} = 1 + 2z + 3z^2 + 4z^3 + \cdots$$

Factorial Notation:

A product of the form $x(x-1)(x-2)\ldots(x-r+1)$ is denoted by $[x]^r$ and is called a factorial.

Thus

$[x] = x$

$[x]^2 = x(x-1)$

$[x]^3 = x(x-1)(x-2)$ and so on.

The operator Δ is used similar to D, when a polynomial is written in factorial notation.

For example

$\Delta[x] = 1$

$\Delta[x]^2 = 2[x]$, $\Delta^2[x]^2 = \Delta 2[x] = 2$

$\Delta[x]^3 = 3[x]^2$, $\Delta^2[x]^3 = 6[x]$, $\Delta^3[x]^3 = 6$.

Example 4:- Solve $y_{n+2} - 4y_n = n^2 + n - 1$

Solution:-

Given equation is $(E^2 - 4)y_n = n^2 + n - 1$

The auxiliary equation $m^2 - 4 = 0$, $\therefore$ $m = \pm 2$

$$C.F = c_1 (2)^n + c_2(-2)^n$$

$$P.I = \frac{1}{E^2 - 4}(n^2 + n - 1)$$

$$= \frac{1}{(1+\Delta)^2 - 4}[n(n-1) + 2n - 1]$$

$$= \frac{1}{\Delta^2 + 2\Delta - 3}([n]^2 + 2[n] - 1) = -\frac{1}{3}\left[1 - \left(\frac{2}{3}\Delta + \frac{\Delta^2}{3}\right)\right]^{-1}\{[n]^2 + 2[n] - 1\}$$

$$= -\frac{1}{3}\left[1 + \left(\frac{2}{3}\Delta + \frac{\Delta^2}{3}\right) + \left(\frac{2}{3}\Delta + \frac{\Delta^2}{3}\right)^2 + \cdots\right]\{[n]^2 + 2[n] - 1\}$$

$$= -\frac{1}{3}\left\{1+\frac{2}{3}\Delta+\frac{7}{9}\Delta^2+\cdots\right\}\{[n]^2+2[n]-1\} = -\frac{1}{3}\{[n]^2+2[n]-1+\frac{2}{3}(2[n]+2)+\frac{7}{9}\times 2\}$$

$$= -\frac{1}{3}\left\{[n]^2+\frac{10}{3}[n]+\frac{17}{9}\right\} = -\frac{n^2}{3}-\frac{7}{9}n-\frac{17}{27}$$

Hence the complete solution is

$$y_n = -\frac{n^2}{3}-\frac{7}{9}n-\frac{17}{27}\,.$$

Example 6: Solve $y_{n+2} + 5y_{n+1} + 6y_n = n + 2^n$

Solution: Given equation in symbolic form is

$$(E^2 + 5E + 6)y_n = n + 2^n$$

Auxiliary equation $c_1 2^n + c_2(-2)^n$ is

$$m^2 + 5m + 6 = 0$$

Roots are -2, -3.

CF =

$$\text{P.I.} = \frac{1}{E^2+5E+6}(n+2^n)$$

$$= \frac{1}{E^2+5E+6}n + \frac{1c1(-3)n+c2(-2)n}{E^2+5E+6}2^n$$

$$= \frac{1}{(1+\Delta)^2+5(1+\Delta)+6}[n] + \frac{1}{2^2+5\times 2+6}2^n$$

$$= \frac{1}{(\Delta^2+7\Delta+12)}[n] + \frac{1}{20}2^n$$

$$= \frac{n}{12}-\frac{7}{144}+\frac{1}{20}2^n$$

The complete solution is

$$y_n = c_1(-3)^n + c_2(-2)^n + \frac{n}{12}-\frac{7}{144}+\frac{1}{20}2^n$$

Case IV:- When f(n) = a^n F(n), F(n) being polynomial of finite degree in n.

$$\text{P.I.} = \frac{1}{\phi(E)}a^nF(n)$$

$$= a^n\frac{1}{\phi(aE)}F(n)$$

Now F(n) being a polynomial in n, proceed as in case III.

Example 6:- Solve $y_{n+2} - 2y_{n+1} + y_n = n^2 \,.\, 2^n$

Solution:- Given equation is

$$(E^2 - 2E + 1)y_n = n^2 \,.\, 2n$$

C.F = $c_1 + c_2 n$

$$\text{P.I} = \frac{1}{(E-1)^2}2^n.n^2$$

$$= 2^n\frac{1}{(2E-1)^2}n^2$$

$$= 2^n \frac{1}{(1+2\Delta)^2} n^2$$

$$= 2^n (1 + 2\Delta)^{-2} \{n(n-1) + n\}$$

$$= 2^n (1 - 4\Delta + 12 \Delta^2 - \ldots)\{[n]^2 + [n]\}$$

$$= 2n \{[n]^2 + [n] - 4(2[n] + 1) + 12 \times 2\}$$

$$= 2n ([n]^2 - 7[n] + 20) = 2n (n^2 - 8n + 20)$$

Hence the complete solution is

$$y_n = c_1 + c_2 n + 2^n (n^2 - 8n + 20)$$

Exercises:

Solve the following difference equation

1. $y_{n+2} - 5y_{n+1} - 6y_n = 4^n$, $y_0 = 0$, $y_1 = 1$
2. $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$, $y_0 = y_1 = 0$
3. $(4E^2 - 4E + 1)_y = 2^n + 2^{-n}$